Department of Electrical and Electrical Engineering, SEET, FUTA.

Course code/Title: EEE 305/307/351 – Electrical Machines 1 **Session:** 2023/2024 **Date:** 3 May, 2024

Instruction: Answer all questions

1. With the aid of good diagrams and two examples explain what you understand by singly and multiply excited systems (6mks)
2. A 200 kVA rated transformer has a full-load copper loss of 2 kW and iron loss of 1.5 kW. Determine the transformer efficiency at (a) full load (b) half full load 0.85 P.f
3. An 8-pole d. c. generator has a lap-wound armature with 1200 conductors. The useful flux per pole is 20 mWb. Find the torque exerted when a current of 20 A flows in each conductor.

THE FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE
School of Engineering and Engineering Technology

Department of Electrical and Electronics Engineering
B. Eng. 1st Semester 2023/2024 Examination

EEE305/307*/351* – Electrical Machines I

Date: 18/05/2024

Instruction: Answer number 1 and any other three questions **Time: $2\frac{1}{4}$ Hrs**

Question 1

(a) With the aid of good diagrams and two examples explain what you understand by singly and multiply excited systems (6mks)

(b) A 500 kVA rated transformer has a full-load copper loss of 3 kW and iron loss of 2 kW. Determine the transformer efficiency at (a) full load and 0.8 (b) half full load and 0.8 (7 mks)

(c) An 8-pole d. c. generator has a wave-wound armature with 1500 conductors. The useful flux per pole is 20 mWb. Determine (a) the e. m. f. generated when running at 500 rpm, and (b) torque exerted when a current of 20 A flows in each conductor (7 mks)

$T = \frac{P\Phi Z I}{\pi C} =$

Question 2

(a) How can the effect of poor commutation in dc machines be improved? (3 mks)

(b) A 4 kVA 400/200 V, 50 Hz single phase transformer has the following test data:

O. C. test; 200 V, 1 A, 64 W

S. C. test; 15 V, 10 A, 80 W. Determine:

(i) equivalent circuit parameters referred to the L. V. side; and

(ii) secondary load voltage on full load at 0.8 p. f. lagging (12 mks)

Question 3

(a) Explain with appropriate diagrams the two types of transformer construction (5 mks)

(b) A transformer has 600 primary turns and 200 secondary turns. The primary and secondary resistances are 0.25Ω and 0.010 Ω respectively and corresponding leakage reactances are 2Ω and 0.08Ω respectively. Determine:

(i) the equivalent resistance referred to the primary winding;

(ii) the equivalent reactance referred to the primary winding;

(iii) the equivalent impedance referred to the primary winding;

(iv) the phase angle of the impedance (10 mks)

$\frac{2P\Phi n Z}{C}$

$\frac{P\Phi Z I}{\pi C}$

Question 4

(a) A three-phase transformer has 600 primary turns and 50 secondary turns. If the supply voltage is 5 kV. Find the secondary line voltage on no-load when the windings are connected (a) star-delta, (b) delta-star. (7 mks)

(b) A shunt dc generator supplies a 24 kW load at 200 V through cables of resistance,

Pry a Sec
Star delta
600 50
V= 5KV V2 ?

R = 100 mΩ. If the field winding resistance, R_f = 50 mΩ and the armature resistance, R_a = 5 m Ω, determine (a) the terminal voltage (b) the e.m.f. generated in the armature (8 mks)

Question 5

(a)) With the aid of a clear diagram describe how to determine the polarity of a transformer (5mks)

(b) With the aid of diagrams write short notes on armature reaction in a dc machines.(5mks)

(c) Evaluate the energy stored in the three coils. For the three coils, given that i_1 = 2A, i_2 = 2.5A, i_3 = 3A, L_{11} = 100mH, L_{22} = 250mH, L_{12} = 1.5H, L_{13} =0.5H, L_{23} = 1H and L_{33} = 150mH. Calculate the total energy stored in the coils (5mks)

Question 6

(a) With the aid of clear diagrams/equations explain why electromechanical energy conversion devices with magnetic field as coupling medium are more common than the devices with electric field (7mks). complete

(b) The lifting magnetic system shown in Fig.Q_6 has a square cross-section 6 x 6cm^2. The coil has 300 turns and a resistance of 6Ω. Neglect reluctance of the magnetic core and field fringing in the air-gap. The air gap is initially held at 5mm and a dc source of 240V is connected to the coil. Determine:

(i) The stored field energy (4mks)

(ii) The lifting force (4mks)

240KA.

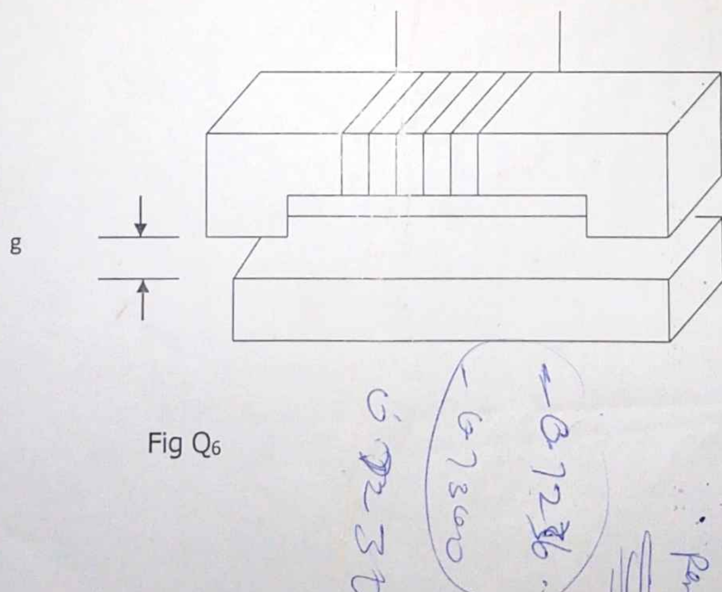

Fig Q_6